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Radiola

OPERATION & SERVICE MANUAL

ESP32 / ESPHome Internet Radio

Motorized Tuning Dial · NFC Record Playback · EM80 Magic Eye · Home Assistant

MODEL	RADIOLA-1
FIRMWARE	radiola1.yaml (ESPHome)
CONTROLLER	ESP32-WROOM

github.com/Christofferfrisk/esphome-devices

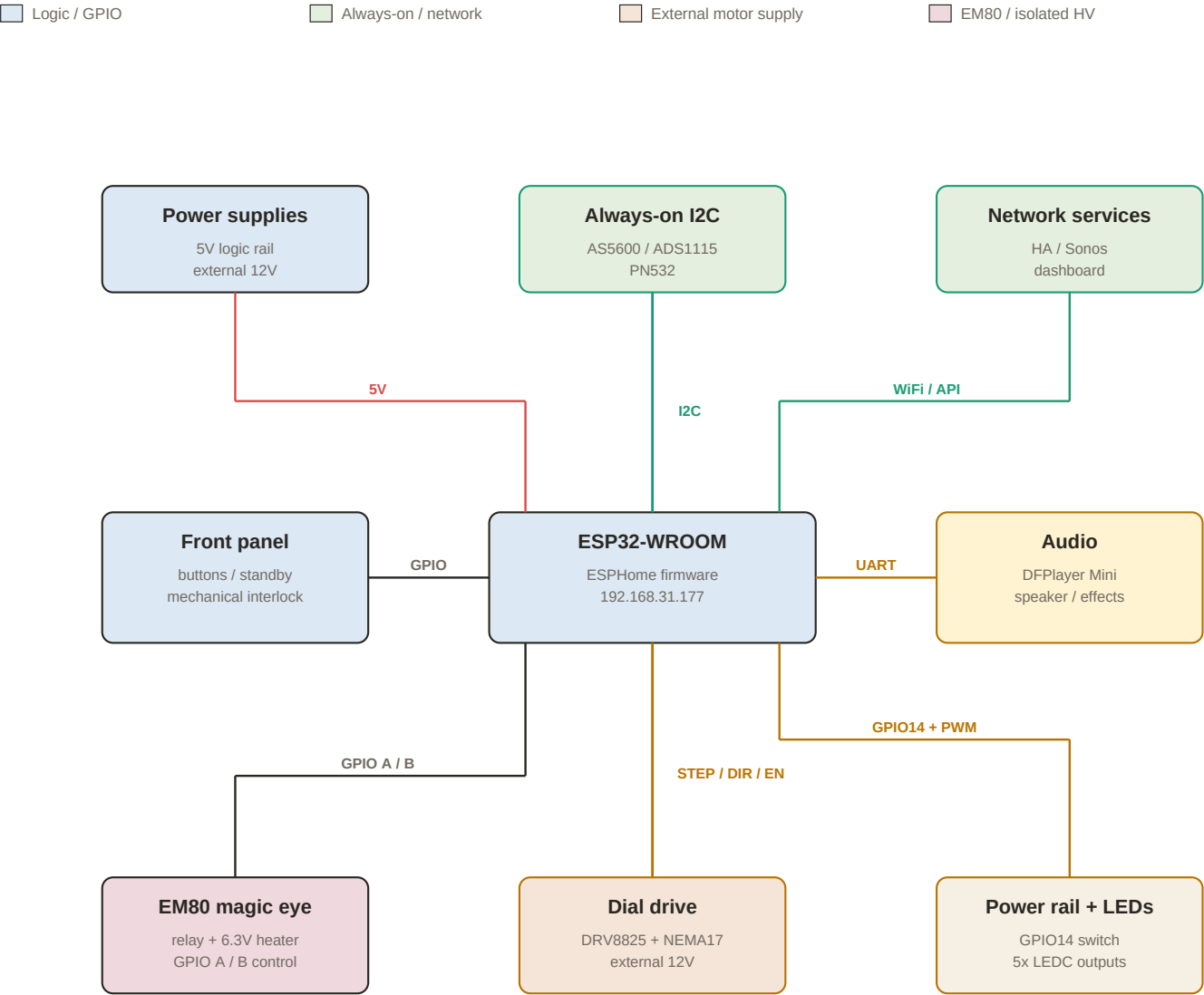
Contains build documentation, wiring reference, and the EM80 magic eye service appendix.

Contents

System block diagram	3
1. Overview	4
2. Subsystems	4
3. Wiring schema	6
4. Power budget (5V logic rail)	6
5. Firmware behavior summary	7
6. Main board circuit wiring reference	8
System wiring at a glance	8
Controls, sensing and audio	9
Power rail, LEDs and standby	10
Dial motor and encoder	11
Appendix - EM80 magic eye reference	12
Pin layout and wiring reference	12
How the EM80 works	13
Complete circuit schematic	14
Build reference	15

Radiola - system block diagram

ESP32 / ESPHome internet radio - motorized dial, NFC, magic eye, Home Assistant



Design notes

- AS5600 remains on the always-on rail so dial position is tracked in standby.
- External 12V feeds the motor and the EM80 relay; LM2596 regulates the heater to 6.3V.
- GPIO14 drives the PMOS rail through a 2N7000 level shifter; do not PWM this output.
- DRV8825 GPIO13/5/15 assignments remain provisional until physical wiring is confirmed.

Radiola — full build documentation

ESP32 / ESPHome motorized internet radio, with encoder-tracked linear dial, NFC “record” playback, DFPlayer sound effects, EM80 magic eye tuning indicator, and Home Assistant / Sonos integration.

1. Overview

Radiola is a DIY internet radio built into a vintage chassis. Tuning works like the original set: a hand-turned knob drives a linear dial pointer along a printed band, stations sit at fixed positions, and static hisses between them. Unlike the original, the ESP32 also knows the pointer's exact position at all times (via a magnetic encoder) and can drive the pointer itself with a small stepper motor — enabling soft mechanical limits, snap-to-station, seek, a standby “parking” motion, and remote tuning from Home Assistant.

2. Subsystems

ESP32-WROOM (main controller)

Runs ESPHome firmware (radiola1.yaml). Talks to all sensors and actuators locally over I2C/UART/GPIO, and to Home Assistant over WiFi via the ESPHome native API. Static IP 192.168.31.177.

AS5600 magnetic rotary encoder (I2C, 0x36)

Absolute single-turn angle sensor reading a magnet on the dial mechanism. Polled every 50 ms; wraparound between samples is accumulated into a multi-turn position (persisted to flash) so the firmware always knows exactly where the pointer is, even across power cycles and standby. Must be powered from the always-on logic rail, not any switched supply, so tracking continues in standby.

ADS1115 ADC (I2C, 0x48)

Reads the volume potentiometer on channel A1. The former dial potentiometer on A0 was retired once the AS5600 took over tuning — channel A0 is currently free.

PN532 NFC reader (I2C)

Detects tagged “records” placed on the radio. In LP mode, a recognized tag triggers album playback on the Sonos via a Home Assistant service call.

DFPlayer Mini (UART)

Plays local WAV files from a microSD card: startup/shutdown chimes, station-tuning static (volume-scaled to the knob and to distance-from-station), and seek sounds. Actual music playback happens on a Sonos speaker via Home Assistant, not through this module.

DRV8825 stepper driver + NEMA17 motor (linear dial)

Drives the physical pointer along the dial cord. EN is held low (driver disabled, knob free-spinning) during normal hand tuning; the firmware energizes it only near the calibrated soft endstops (spring-back — assists turning back in range, resists going further out, never locks the knob) or during a commanded move (seek, snap-to-station, standby park, LP progress). Powered from its own dedicated motor supply, isolated from the logic rail, since stepper current transients would otherwise disturb the 5V rail.

5x status LEDs (LEDC + MOSFET outputs)

Dimmable via PWM. Used for the startup/shutdown “wake-up” sweep animation and as calibration feedback (blinking during the limit-setting sequence).

4x latching buttons + standby contact plate (GPIO)

Mechanically interlocked piano-key buttons (pressing one releases any other). Button functions: mode select (radio/LP), seek, and the 5-press calibration trigger. A separate spring contact plate detects the physical standby position and cuts the power rail via a GPIO-driven switch (converted from an earlier dimmable-light implementation, since PWMing a power rail risked chopping the DFPlayer's supply).

EM80 magic eye tuning indicator

A vacuum magic eye tube (EM80/6BR5) whose shadow narrows as the dial approaches a station, driven from the same tuning-distance signal used for the static swim effect. GPIO A controls a 3.3V-compatible relay module which switches 12V to both the MAX1771 and an LM2596 buck. The buck is adjusted to 6.3V for the heater. GPIO B drives a PC817 at about 5 kHz to vary the shadow. Anode/target voltage comes from the MAX1771 configured for roughly 80V. Full wiring and safety notes are reproduced in the reference schematic at the end of this document.

Home Assistant + Sonos + dashboard card

HA automations watch the ESP32's published sensors (Dial Mode, Radiola Preload Station, LP Progress) to start/stop Sonos playback, preload the next station's playlist muted while the dial is still in the static gap, and ramp volume on arrival. A custom Lovelace card renders a vintage dial face with a live needle mirroring the physical pointer, a mode badge, station/track label, and play/pause control.

Power domains at a glance

Always-on logic

5V rail

ESP32, AS5600, ADS1115

Switched 5V

GPIO14 load switch

DFPlayer Mini and LEDs

External supplies

12V motor + EM80

6.3V heater via LM2596

3. Wiring schema

Literal ESP32 pin assignments for each subsystem, from radiola1.yaml. DRV8825 pins are substitution placeholders — confirm and update before wiring. For the EM80 magic eye's own pin-out and wire colors, see the reference schematic appended at the end.

ESP32 pin	Function	Connects to	Notes
GPIO21	I2C SDA	AS5600, ADS1115, PN532	Shared bus
GPIO22	I2C SCL	AS5600, ADS1115, PN532	Shared bus
GPIO17	UART TX	DFPlayer Mini RX	
GPIO16	UART RX	DFPlayer Mini TX	
GPIO27	LEDC PWM	LED 1 (via MOSFET)	
GPIO25	LEDC PWM	LED 2 (via MOSFET)	
GPIO32	LEDC PWM	LED 3 (via MOSFET)	
GPIO33	LEDC PWM	LED 4 (via MOSFET)	
GPIO26	LEDC PWM	LED 5 (via MOSFET)	
GPIO14	Digital out	2N7000 gate (PMOS level shift)	GPIO HIGH enables; never PWM
GPIO23	Digital in, pull-up	Button 1	Mechanically interlocked
GPIO12	Digital in, pull-up	Button 2	Avoid — ESP32 boot-strapping pin
GPIO18	Digital in, pull-up	Button 3	
GPIO19	Digital in, pull-up	Button 4	
GPIO4	Digital in, pull-up	Standby contact plate	Plate on GND = active; away = standby
GPIO13*	Step pulse	DRV8825 STEP	*substitution placeholder
GPIO5*	Direction	DRV8825 DIR	*substitution placeholder
GPIO15*	Enable, active-low	DRV8825 EN	*provisional; add 10k EN pull-up to 3.3V for coils-off boot
GPIO A (TBD)	Digital out	EM80 relay module S	3.3V-compatible input
GPIO B (TBD)	PWM ~5kHz	PC817 LED via 470R	C->GND; E->Pin 1

4. Power budget (5V logic rail)

Estimated current draw for everything sharing the 5V logic supply. The NEMA17 motor and the EM80's heater/HV supply are deliberately kept off this rail (see Subsystems).

Component	Typical	Peak	Notes
ESP32-WROOM	~80 mA	500 mA	WiFi TX bursts spike hard and briefly
DFPlayer Mini	40–60 mA	300–500 mA	Volume- and speaker-load dependent
PN532	~20 mA	100–150 mA	Peaks only while energizing a tag
5x LEDs	20–40 mA	75–100 mA	All five lit (startup sweep) is the peak case
AS5600	6.5 mA	~7 mA	Always on, incl. in standby
NEMA17 via DRV8825*	external 12V	Vref-limited	Separate motor supply - excluded
ADS1115 (5V VDD)	0.15 mA	0.26 mA	5V rail; I2C must remain 3.3V-safe
EM80 magic eye*	~0.3 A @12V	0.5 A	Relay-switched 12V - excluded from total
Total	~190 mA	~1.2 A	≈ 1 W typical / ≈ 6 W peak — supply: 5V/2A min

* External 12V feeds NEMA17/DRV8825 and the separately switched EM80 branch.
The LM2596 regulates the EM80 heater to 6.3V; external loads are excluded from the 5V total.

5. Firmware behavior summary

- **Soft endstops:** dial_min/dial_max calibrated by hand; the motor springs the pointer back toward the safe span near either limit rather than locking it in place.
- **Stations:** four positions at 20/40/60/80% of the usable dial span, each with a capture zone.
- **Snap-to-station:** once the hand-turned dial settles near a station, the motor centers it.
- **Seek:** sweeps to the next station with static hissing until arrival.
- **Static “swim”:** hiss volume scales with distance from the nearest station and the volume knob.
- **Preload:** while tuning through static, a sensor names the station being approached (direction-aware, with jitter hysteresis) so Home Assistant can start that playlist muted before arrival.
- **Standby park:** the pointer glides to the left end while the shutdown chime plays, then the power rail cuts.
- **Calibration:** 5x press of Button 4 enters calibration; Button 3 captures the left then right physical limits.
- **Failsafe:** if the encoder goes silent for more than 1 second, the motor is disabled and moves are locked out until recalibrated.

Source: [esphome-devices/radiola](#) ([radiola1.yaml](#), [README.md](#), [homeassistant/](#)). Values marked as estimates should be verified against real hardware. The EM80 pin-out, circuit schematic, and parts list on the following pages are reproduced from the supplied reference document.

Automatic standby

Timeout	After 3 minutes without physical interaction, the Radiola enters standby.
Sequence	Shutdown sound, LED fade, Sonos stop, dial park, motor release, rail off.
Reset / wake	Dial, volume, Radio/LP buttons, or NFC activity reset the timer and wake it.

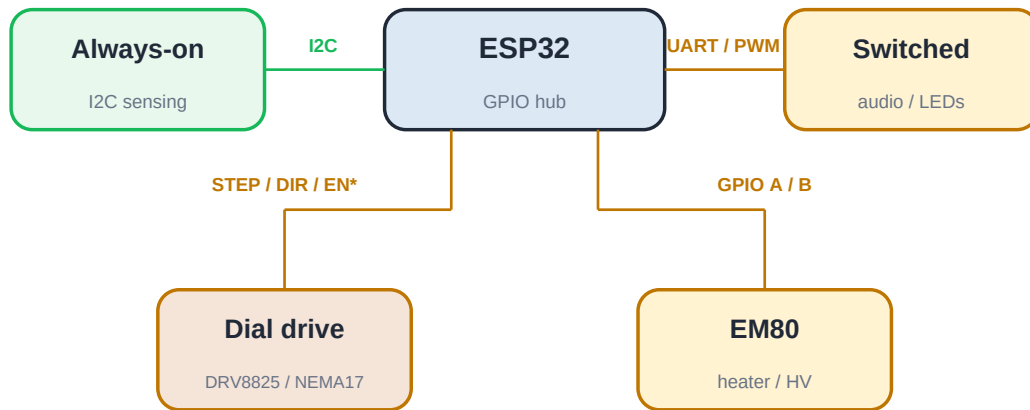
Motor-driven snap and park movements do not count as activity. The physical standby plate remains authoritative.

Pre-power checklist

- ❑ I2C pull-ups go to 3.3V only; remove any ADS1115 pull-ups tied to its 5V VDD.
- ❑ DRV8825 VMOT is on the external 12V supply with 100uF at the carrier.
- ❑ DRV8825 EN has a 10k pull-up to 3.3V so the coils remain off during ESP32 boot.
- ❑ Verify the GPIO14 level-shifted PMOS rail is off before enabling connected loads.
- ❑ Discharge and meter the EM80 HV output before touching or changing the circuit.

1. System wiring at a glance

Follow named connectors; power rails and signal roles use a consistent colour key.

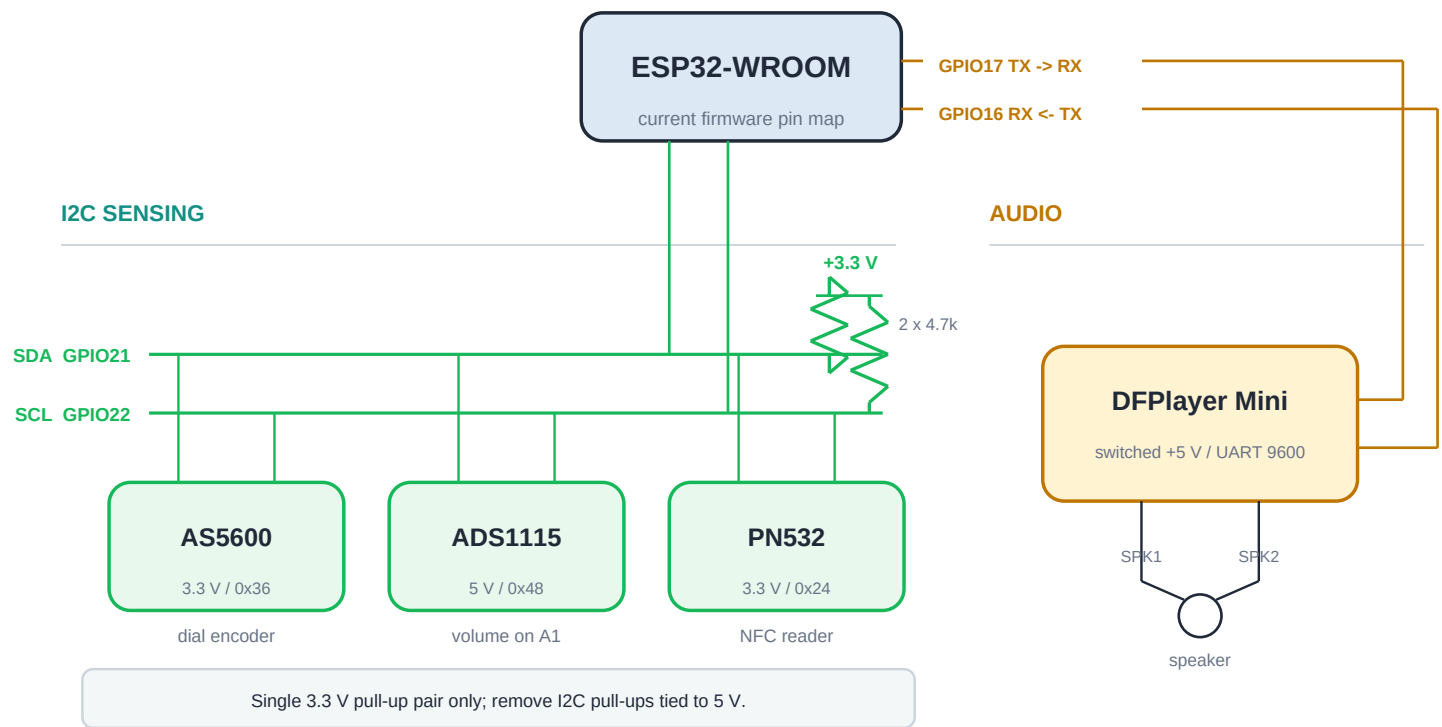


* DRV8825 pins are firmware placeholders - confirm physical wiring.

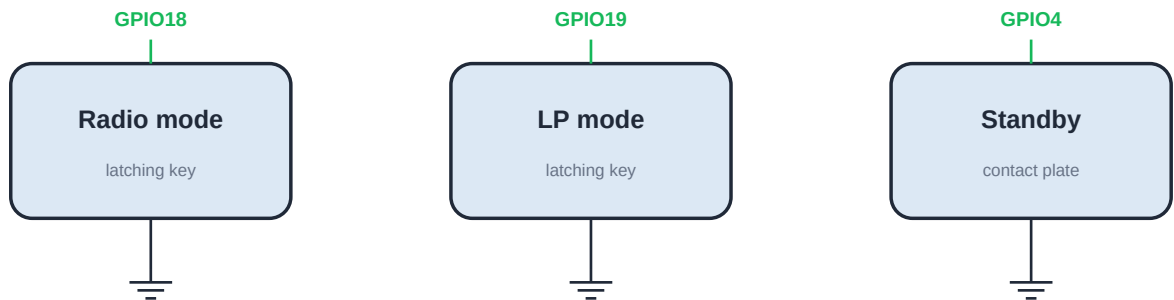
Role	Colour	Meaning
Power	Red +5 V / amber HV	Supply rails; connect only to the labelled rail
ESP / signal	Green / charcoal	GPIO, I2C, UART and control connections
Ground	Charcoal	All low-voltage modules share common GND

2. Controls, sensing and audio

Current low-voltage wiring, grouped by function with no shared signal crossings.



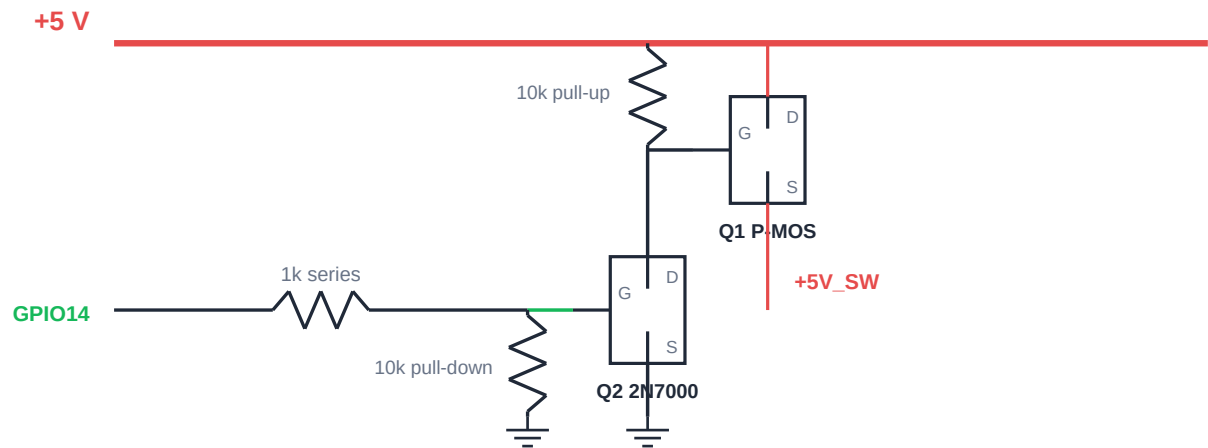
FRONT-PANEL INPUTS



Module / input	Power	Interface	Operational note
AS5600	always-on 3.3 V	I2C 0x36	tracked in standby
ADS1115	always-on 5 V	I2C 0x48 / A1	I2C stays 3.3 V-safe
PN532	3.3 V	I2C 0x24	reader in I2C mode
DFPlayer Mini	switched +5 V	UART 9600	TX/RX crossed
Radio / LP / standby	ESP32 pull-ups	GPIO18 / 19 / 4	contacts close to GND

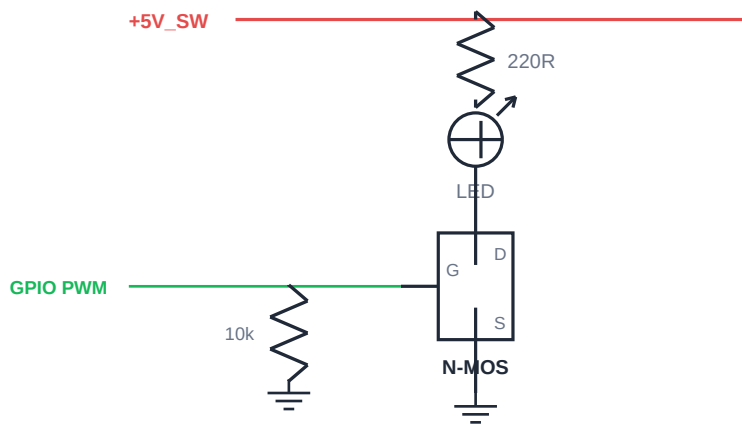
3. Power rail, LEDs and standby

A compact connected schematic for the switched rail and one repeated LED driver stage.



GPIO14 HIGH turns Q2 on, pulls Q1 gate LOW and enables +5V_SW; GPIO LOW leaves the rail OFF.

LED low-side driver - repeat this stage x5

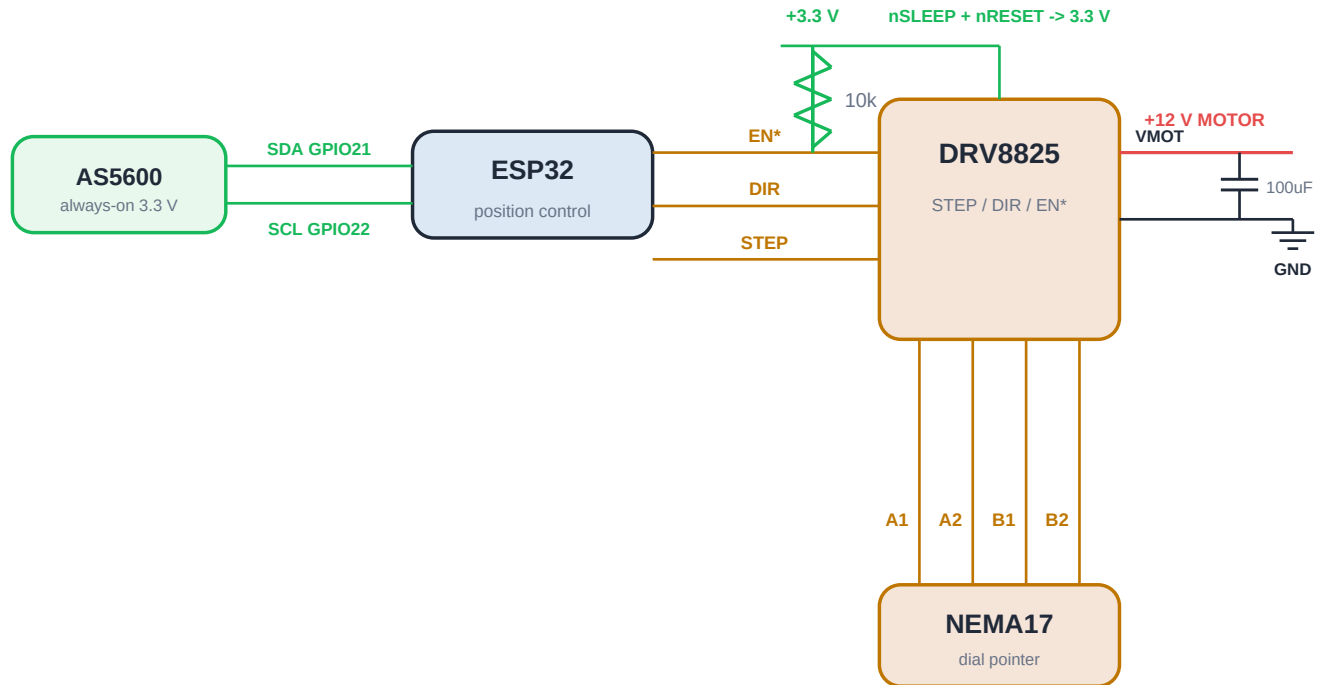


LED1 GPIO27 / LED2 GPIO25 / LED3 GPIO32 / LED4 GPIO33 / LED5 GPIO26
Each channel gets its own 220R resistor, N-MOS stage and 10k gate pull-down.

Circuit	ESP32 control	Required note
Switched rail	GPIO14	HIGH enables via Q2 level shifter; never PWM
LED channels 1-5	GPIO27 / 25 / 32 / 33 / 26	One MOSFET stage per LED
Standby plate	GPIO4	Plate to GND = active; open = standby

4. Dial motor and encoder

Keep motor power separate from ESP32 logic; confirm the driver's ground arrangement.



DRV8825 pin	Firmware mapping	Build requirement
STEP	GPIO13*	Provisional - verify before flashing
DIR	GPIO5*	Provisional - verify before flashing
EN (active low)	GPIO15*	Add 10k pull-up to 3.3 V for coils-off boot
VMOT / GND	external 12 V	100 uF electrolytic directly at the driver
Logic inputs	3.3 V signals	No VDD pin; ESP32 and driver share GND
nSLEEP / nRESET	tie high together	Required for STEP pulses to operate

Safety

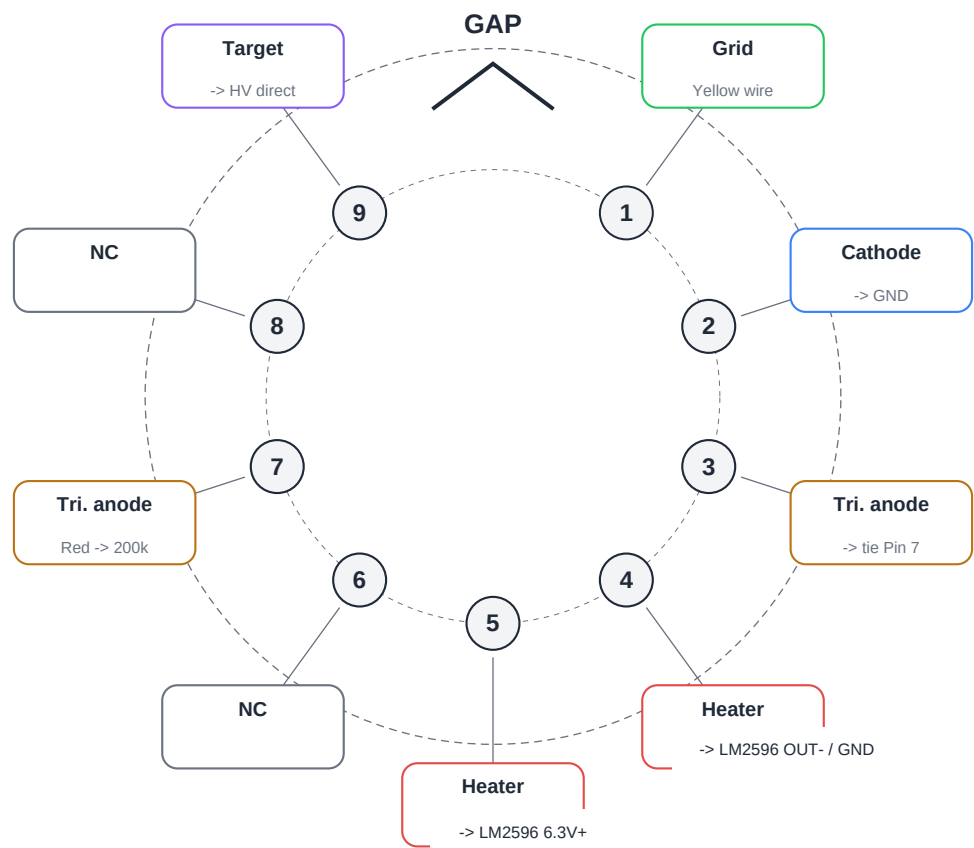
Set the DRV8825 current limit conservatively. Never connect or disconnect the motor while powered.

EM80 Magic Eye Tube

ESP32 Radiola Project — Wiring Reference

Pin Layout — Bottom View (looking at pins)

Pins numbered clockwise from gap (standard Noval B9A)



Pin Reference Table

Pin	Function	Connection	Wire
1	Grid (control input)	PC817 emitter + 1M to GND	
2	Cathode	GND	Black
3	Triode anode (internal)	Tie to Pin 7 externally	-
4	Heater	LM2596 OUT- / common GND	
5	Heater	LM2596 6.3V+	
6	No connection	-	-
7	Triode anode	200k resistor to HV+	Red
8	No connection	-	-
9	Target anode (display)	HV+ direct (~80V)	-

The LM2596 supplies the EM80/6BR5 heater with a regulated 6.3V across Pins 5 and 4.

How the EM80 Works

Understanding each element of the tube

Heater (Pin 4 & 5)

A thin filament inside the tube, like a light bulb element. Put voltage across these pins and the cathode heats up so it can emit electrons. Takes 10-15 seconds to warm up.

Cathode (Pin 2)

The electron source. When heated by the filament, it emits electrons toward the target. Connected directly to GND.

Target Anode (Pin 9)

The fluorescent screen that glows green. Electrons from the cathode hit this surface. Gets HV directly. The shadow appears where electrons are deflected away.

Grid (Pin 1)

The control input. A small voltage change here (0V to -14V) controls how much the triode conducts. This is where the ESP32 PWM signal controls the shadow.

Triode Anode (Pin 3 & Pin 7)

The amplifier output. Connected to HV through a 200k load resistor. The voltage here determines the shadow position on the display.

How Grid Self-Bias Works

The grid (Pin 1) is connected to GND through a large resistor (1M or more). Inside the tube, some electrons from the hot cathode land on the grid. This tiny current flows through the resistor to ground. By Ohm's law, even a few microamps through 1M creates several volts of drop, making the grid negative relative to ground. This is called 'grid leak bias' and it happens naturally.

When the Q2 optocoupler turns on, its output transistor clamps Pin 1 close to GND, overriding the self-bias and forcing the grid toward 0 V. PWM duty controls how long the clamp is active, varying the average grid voltage and shadow.

Shadow Behavior

PWM low (0%) -> Q2 off -> Grid self-biases negative
 -> Shadow closes -> MORE green glow = TUNED

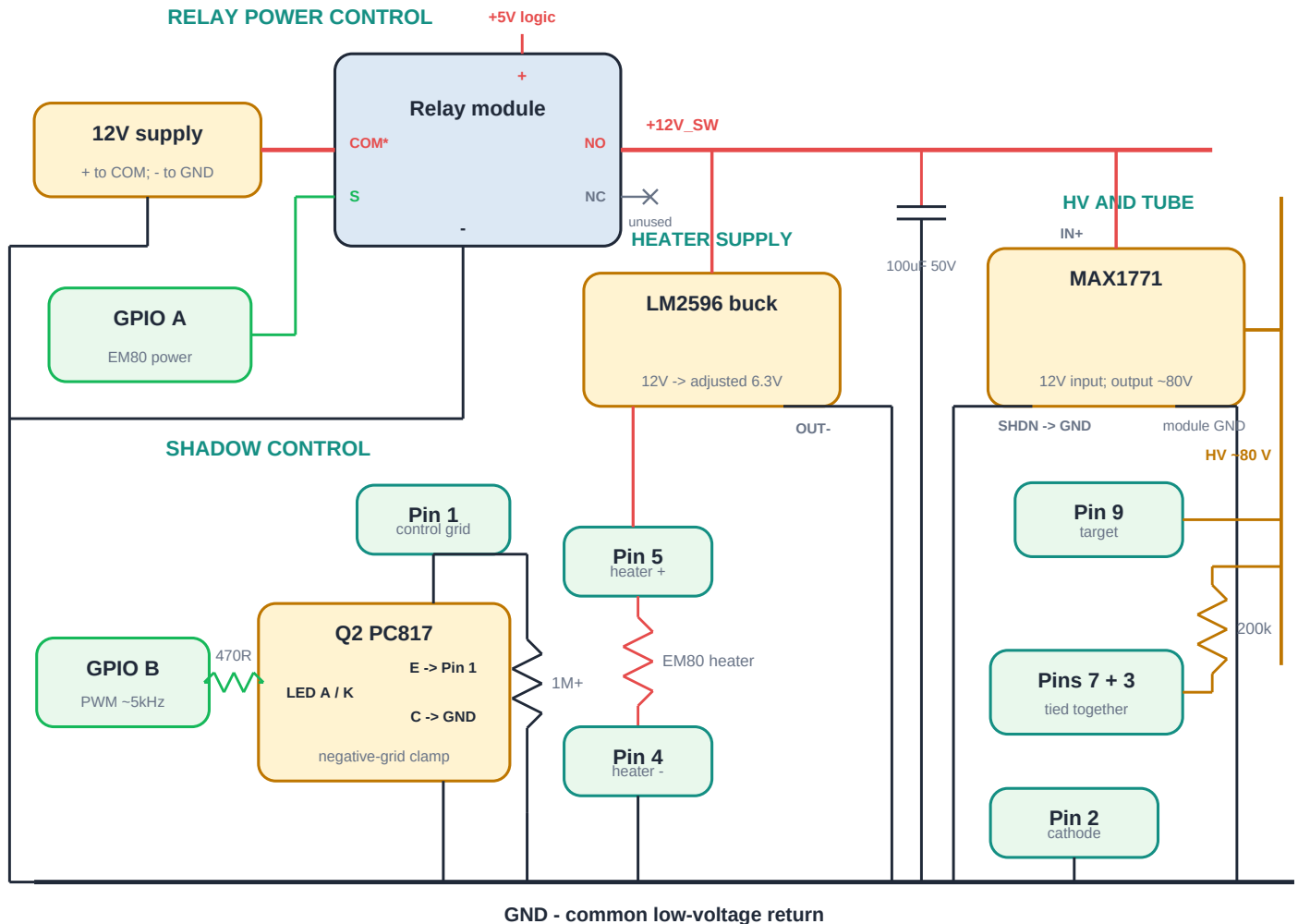
PWM high (100%) -> Q2 on -> Grid clamped to 0V
 -> Shadow opens -> LESS green glow = BETWEEN

Note on bias resistor value:

Start with 1M. If shadow range is too small, add more 1M resistors in series.
2-5M gives more range. Larger resistor = more negative bias = wider shadow sweep.

Complete Circuit Schematic

ESP32 Radiola - relay power, regulated heater, shadow and HV wiring



Important Q2 orientation

PC817 collector -> GND; emitter -> Pin 1. When its LED is off, the 1M resistor permits negative grid self-bias.
Relay markings: S=GPIO A, +=5V, -=GND; COM* is the unmarked centre terminal between NC and NO.

Section	Connection	Required note
Power	+12V -> relay COM*; NO -> +12V_SW	S=GPIO A; +=5V; -=GND
Heater	+12V_SW -> LM2596 -> Pins 5 / 4	Adjust and meter 6.3V before fitting tube
Shadow	Pin 1 -> 1M + PC817 emitter	GPIO B drives opto LED through 470R
HV	Pin 9 direct; Pins 7+3 via 200k	Adjust MAX1771 output to about 80 V
MAX1771	IN+ -> +12V_SW; SHDN -> GND	100uF decoupling at module input

HIGH VOLTAGE

Disconnect power, discharge the HV output and verify 0 V before touching the circuit.

Build Reference

Parts list, connections, and safety

Parts List

Component	Value / Type	Qty	Notes
EM80 (6BR5)	Magic eye tube	1	Noval 9-pin base
LM2596 module	Adjustable buck	1	Set to 6.3V before fitting tube
Relay module	S / + / - input	1	NC / COM* / NO; COM* is unmarked
PC817	Optocoupler	1	Q2 shadow clamp; C->GND, E->Pin 1
MAX1771 module	5-12V to HV DC	1	Powered from switched 12V; set ~80V
Resistor	200k	1	Triode load resistor
Resistor	1M or more	1+	Grid self-bias
Resistor	470R	1	PC817 LED current limiting
Capacitor	100uF 50V	1	Switched 12V input decoupling
Noval socket	B9A 9-pin	1	For the EM80 tube

Control and Power Connections

Source	Function	Connection
GPIO A	Digital output	Relay module S terminal
GPIO B	PWM ~5kHz	PC817 LED through 470R (shadow control)
+12V	External supply	Unmarked centre (COM*); NO becomes +12V_SW
GND	Common return	ESP32, relay -, LM2596 and MAX1771 grounds

Breadboard to Tube Socket (6 wires)

Source	Function	Connection
Pin 1	Grid (yellow)	PC817 emitter + 1M junction
Pin 2	Cathode (black)	Common GND
Pin 4	Heater - (black)	LM2596 OUT- / common GND
Pin 5	Heater + (blue)	LM2596 OUT+ adjusted to 6.3V
Pin 7	Triode (red)	200k to HV+; jumper Pin 3 to Pin 7
Pin 9	Target	HV+ direct from MAX1771

Safety Notes — HIGH VOLTAGE

- The HV module may exceed 200V; this build is adjusted to roughly 80V.
- NEVER touch the circuit while powered on.
- After powering off, wait at least 60 seconds before touching the HV side.
- Use a multimeter to verify 0V on HV output before working on it.
- Keep HV wiring physically separated from low voltage breadboard.
- Solder HV connections on perfboard, not breadboard.
- Insulate all exposed HV connections with heat shrink tubing.
- Use only one hand when measuring or probing the HV side.